Variation of Parameters

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The Wronskian**

For $y_1 = e^x$, $y_2 = e^{2x}$ (solutions of $y'' - 3y' + 2y = 0$), compute the Wronskian $W = y_1y_2' - y_1'y_2$.

$y_1' = e^x$, $y_2' = 2e^{2x}$. $W = e^x(2e^{2x}) - e^x(e^{2x}) = 2e^{3x} - e^{3x} = e^{3x}$.

2 **[Foundational]** • **Target: The Wronskian**

For $y_1 = \cos x$, $y_2 = \sin x$, compute W.

$y_1' = -\sin x$, $y_2' = \cos x$. $W = \cos x\cos x - (-\sin x)\sin x = \cos^2 x + \sin^2 x = 1$.

3 **[Foundational]** • **Target: The Wronskian**

For $y_1 = e^{3x}$, $y_2 = xe^{3x}$ (solutions of a repeated-root equation), compute W.

$y_1' = 3e^{3x}$, $y_2' = e^{3x} + 3xe^{3x}$. $W = e^{3x}(e^{3x} + 3xe^{3x}) - 3e^{3x}(xe^{3x}) = e^{6x} + 3xe^{6x} - 3xe^{6x} = e^{6x}$.

4 **[Foundational]** • **Target: The variation of parameters formula**

State the variation of parameters formula for a particular solution of $y'' + p(x)y' + q(x)y = g(x)$, given independent solutions y_1, y_2 of the homogeneous equation with Wronskian W.

$y_p = -y_1\int \frac{y_2 g}{W}\,dx + y_2\int \frac{y_1 g}{W}\,dx$.

5 **[Foundational]** • **Target: The variation of parameters formula**

Explain why the ODE must first be written in standard form $y'' + p(x)y' + q(x)y = g(x)$ (leading coefficient 1) before applying the variation of parameters formula.

The formula's Wronskian and integrals are derived under the assumption that $g(x)$ is exactly the right side once the leading coefficient of y'' is 1. If the equation is given as $a(x)y'' + b(x)y' + c(x)y = f(x)$ with $a(x) \neq 1$, using $f(x)$ directly in place of $g(x)$ would introduce an extra factor of $a(x)$ and give a wrong particular solution — the equation must first be divided by $a(x)$, giving $g(x) = f(x)/a(x)$.

6 **[Foundational]** • **Target: A forcing term with no undetermined-coefficients trial form**

Solve $y'' + y = \tan x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$ using variation of parameters.

Here $y_1 = \cos x$, $y_2 = \sin x$, $W = 1$, $g = \tan x$. Since $\int \cos x\tan x\,dx = \int \sin x\,dx = -\cos x$ and $\int \sin x\tan x\,dx = \int \sec x\,dx - \int \cos x\,dx = \ln|\sec x + \tan x| - \sin x$, we get $y_p = -\cos x[\ln|\sec x + \tan x| - \sin x] + \sin x(-\cos x) = -\cos x\ln|\sec x + \tan x|$.

General solution: $y = C_1\cos x + C_2\sin x - \cos x\ln|\sec x + \tan x|$.

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7 [Foundational] • Target: A forcing term with no undetermined-coefficients trial form

Solve $y'' + y = \sec x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$ using variation of parameters.

$y_1 = \cos x$, $y_2 = \sin x$, $W = 1$, $g = \sec x$. $\frac{y_2 g}{W} = \frac{\sin x}{\cos x} = \tan x$, so $\int = -\ln|\cos x|$. $\frac{y_1 g}{W} = \frac{\cos x}{\cos x} = 1$, so $\int = x$. Then $y_p = -\cos x(-\ln|\cos x|) + \sin x(x) = \cos x \ln|\cos x| + x\sin x$. General solution: $y = C_1\cos x + C_2\sin x + \cos x\ln|\cos x| + x\sin x$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: A repeated root, with forcing dividing by x

Solve $y'' - 2y' + y = \frac{e^x}{x}$ $(x > 0)$ using variation of parameters.

$r^2 - 2r + 1 = (r-1)^2 = 0$, so $y_1 = e^x$, $y_2 = xe^x$, and $W = e^{2x}$. With $g = e^x/x$: $\frac{y_2 g}{W} = 1$ so $\int = x$; $\frac{y_1 g}{W} = \frac{1}{x}$ so $\int = \ln x$. Then $y_p = -e^x(x) + xe^x(\ln x) = xe^x(\ln x - 1)$. Absorbing the $-xe^x$ term into the arbitrary constant on y_2, the general solution is $y = C_1e^x + C_2xe^x + xe^x\ln x$.

9 [Intermediate] • Target: A repeated root, with forcing dividing by x

Solve $y'' - 4y' + 4y = \frac{e^{2x}}{x^2}$ $(x > 0)$ using variation of parameters.

$r^2 - 4r + 4 = (r-2)^2 = 0$, so $y_1 = e^{2x}$, $y_2 = xe^{2x}$, $W = e^{4x}$. With $g = e^{2x}/x^2$: $\frac{y_2 g}{W} = \frac{1}{x}$, integral $= \ln x$; $\frac{y_1 g}{W} = \frac{1}{x^2}$, integral $= -\frac{1}{x}$. Then $y_p = -e^{2x}\ln x + xe^{2x}\left(-\frac{1}{x}\right) = -e^{2x}\ln x - e^{2x}$. Absorbing the constant $-e^{2x}$ term into C_1, the general solution is $y = C_1e^{2x} + C_2xe^{2x} - e^{2x}\ln x$.

10 [Intermediate] • Target: Comparing with undetermined coefficients

Solve $y'' - y = e^{2x}$ by variation of parameters, then check the particular solution using undetermined coefficients.

$y_1 = e^x$, $y_2 = e^{-x}$, $W = -2$. With $g = e^{2x}$: $\frac{y_2 g}{W} = -\frac{1}{2}e^x$, integral $-\frac{1}{2}e^x$; $\frac{y_1 g}{W} = -\frac{1}{2}e^{3x}$, integral $-\frac{1}{6}e^{3x}$. So $y_p = -e^x(-\frac{1}{2}e^x) + e^{-x}(-\frac{1}{6}e^{3x}) = \frac{1}{2}e^{2x} - \frac{1}{6}e^{2x} = \frac{1}{3}e^{2x}$. Check by undetermined coefficients: try Ae^{2x} (no resonance, since 2 is not a root of $r^2 - 1 = 0$): $4A - A = 1 \Rightarrow A = \frac{1}{3}$. Matches. General solution: $y = C_1e^x + C_2e^{-x} + \frac{1}{3}e^{2x}$.

11 [Intermediate] • Target: Comparing with undetermined coefficients

Solve $y'' - y = 4$ (a constant forcing) by variation of parameters, and check against the obvious undetermined-coefficients guess $y_p = A$.

$y_1 = e^x$, $y_2 = e^{-x}$, $W = -2$, $g = 4$. $\frac{y_2 g}{W} = -2e^{-x}$, integral $2e^{-x}$; $\frac{y_1 g}{W} = -2e^x$, integral $-2e^x$. So $y_p = -e^x(2e^{-x}) + e^{-x}(-2e^x) = -2 - 2 = -4$. Undetermined coefficients: try $y_p = A$, giving $-A = 4 \Rightarrow A = -4$. Matches. General solution: $y = C_1e^x + C_2e^{-x} - 4$.

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12 **[Intermediate]** • **Target: When to prefer each method**

For $y'' + 4y = x^3 - 2x$, which method is more efficient: undetermined coefficients or variation of parameters? Explain.

Undetermined coefficients is far more efficient here: the forcing is a plain polynomial (no resonance issues), so a simple trial $y_p = Ax^3 + Bx^2 + Cx + D$ solves it by matching coefficients — no integration required. Variation of parameters would work too, but requires computing W and two integrals unnecessarily.

13 **[Intermediate]** • **Target: When to prefer each method**

For $y'' + y = \csc x$ (for $0 < x < \pi$), which method must be used, and why?

Variation of parameters is required. $\csc x = 1/\sin x$ is not a polynomial, exponential, sine/cosine, or a finite product of these, so undetermined coefficients has no valid trial form; only variation of parameters, which places no restriction on the forcing term, can produce a particular solution.

14 **[Intermediate]** • **Target: Solving $y''+y=\csc x$ in full**

Complete the solution of $y'' + y = \csc x$ ($0 < x < \pi$) started above, using $y_1 = \cos x$, $y_2 = \sin x$, $W = 1$.

$\frac{y_2 g}{W} = \frac{\sin x}{\sin x} = 1$, so $\int = x$. $\frac{y_1 g}{W} = \frac{\cos x}{\sin x} = \cot x$, so $\int = \ln|\sin x|$. Then $y_p = -\cos x(x) + \sin x \ln|\sin x| = -x\cos x + \sin x \ln|\sin x|$. General solution: $y = C_1 \cos x + C_2 \sin x - x\cos x + \sin x \ln|\sin x|$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Solving $y''+y=\csc x$ in full** • **Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations**

Verify (by checking the coefficient of the leading term only) that $y_p = -x\cos x$ contributes a term of the correct order when substituted into $y'' + y$.

For $y_p = -x\cos x$: $y_p' = -\cos x + x\sin x$, $y_p'' = \sin x + \sin x + x\cos x = 2\sin x + x\cos x$. So $y_p'' + y_p = 2\sin x + x\cos x - x\cos x = 2\sin x$ — this piece alone contributes $2\sin x$, which combines with the contribution from $\sin x \ln|\sin x|$ to produce the required $\csc x$ on the right side (a full check of the second piece is a longer computation, but this confirms the $-x\cos x$ term is doing real work, not just decoration).

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16 **[Challenging] • Target: Variation of parameters with a shifted independent variable • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients**

For the Cauchy–Euler-type substitution used in some ODE courses, $y_1 = x$, $y_2 = x^{-1}$ solve $x^2y'' + xy' - y = 0$ for $x > 0$. Rewrite this equation in the standard form $y'' + p(x)y' + q(x)y = g(x)$ needed for variation of parameters when the right side is $x^2 \cdot 3x = 3x^3$ (i.e., solve $x^2y'' + xy' - y = 3x^3$).

Dividing by x^2: $y'' + \frac{1}{x}y' - \frac{1}{x^2}y = 3x$, so $g(x) = 3x$ (NOT $3x^3$ — the division by the leading coefficient x^2 is exactly the step the earlier question about standard form warns about).

17 **[Challenging] • Target: Variation of parameters with a shifted independent variable • Cross-Lesson: 1.01 Separable Equations**

Continue the problem above: with $y_1 = x$, $y_2 = x^{-1}$, compute $W = y_1y_2' - y_1'y_2$.

$y_1' = 1$, $y_2' = -x^{-2}$. $W = x(-x^{-2}) - 1(x^{-1}) = -x^{-1} - x^{-1} = -2x^{-1}$.

18 **[Challenging] • Target: Variation of parameters with a shifted independent variable • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor**

Finish solving $x^2y'' + xy' - y = 3x^3$ ($x > 0$) using $y_1 = x$, $y_2 = x^{-1}$, $W = -2x^{-1}$, and $g(x) = 3x$ (from the two questions above).

$\frac{y_2g}{W} = \frac{x^{-1}(3x)}{-2x^{-1}} = -\frac{3x}{2}$, so $\int = -\frac{3x^2}{4}$. $\frac{y_1g}{W} = \frac{x(3x)}{-2x^{-1}} = -\frac{3x^3}{2}$, so $\int = -\frac{3x^4}{8}$. Then $y_p = -x\left(-\frac{3x^2}{4}\right) + x^{-1}\left(-\frac{3x^4}{8}\right) = \frac{3x^3}{4} - \frac{3x^3}{8} = \frac{3x^3}{8}$. General solution: $y = C_1x + C_2x^{-1} + \frac{3x^3}{8}$.

19 **[Challenging] • Target: Variation of parameters with a shifted independent variable • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations**

Check the particular solution $y_p = \frac{3}{8}x^3$ found above by substituting directly into $x^2y'' + xy' - y = 3x^3$.

$y_p' = \frac{9}{8}x^2$, $y_p'' = \frac{9}{4}x$. Then $x^2y_p'' + xy_p' - y_p = x^2\left(\frac{9}{4}x\right) + x\left(\frac{9}{8}x^2\right) - \frac{3}{8}x^3 = \frac{9}{4}x^3 + \frac{9}{8}x^3 - \frac{3}{8}x^3 = \frac{18}{8}x^3 + \frac{9}{8}x^3 - \frac{3}{8}x^3 = \frac{24}{8}x^3 = 3x^3$. Matches.

20 **[Challenging] • Target: A summary comparison • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients**

List two advantages of variation of parameters over undetermined coefficients.

(1) It works for ANY continuous forcing term $g(x)$, not just polynomials/exponentials/sines-cosines and their products — it handles $\tan x$, $\sec x$, $1/x$, $\ln x$, and more. (2) It gives a single unified formula/procedure, with no need to guess a trial form or check for resonance case-by-case.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: The Wronskian Determinant

What is the definition of the Wronskian $W(y_1, y_2)$ of two differentiable functions $y_1(x)$ and $y_2(x)$?

a. $W(y_1, y_2) = y_1y_2' - y_1'y_2 = \det[[y_1, y_2], [y_1', y_2']]$ ✓

b. $W(y_1, y_2) = y_1y_2 + y_1'y_2'$

c. $W(y_1, y_2) = y_1'y_2 - y_1y_2'$

d. $W(y_1, y_2) = \frac{y_1}{y_2}$

The Wronskian is the determinant of the fundamental matrix: $W = \det[[y_1, y_2], [y_1', y_2']] = y_1y_2' - y_1'y_2$.

22 [Foundational] • Target: The Wronskian Determinant

Compute the Wronskian of $y_1 = \cos x$ and $y_2 = \sin x$.

a. 1 ✓

b. 0

c. -1

d. $\cos(2x)$

$W = (\cos x)(\cos x) - (-\sin x)(\sin x) = \cos^2 x + \sin^2 x = 1$.

23 [Foundational] • Target: The Wronskian Determinant

What does $W(y_1, y_2) \neq 0$ guarantee about solutions y_1 and y_2 to a homogeneous linear ODE?

a. **They are linearly independent and form a fundamental set of solutions** ✓

b. They are orthogonal

c. They are periodic

d. One must be zero

If the Wronskian is non-zero at a point in the interval, Abel's theorem guarantees it is non-zero everywhere, establishing linear independence.

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24 [Foundational] • Target: The Variation of Parameters Formula

For the second-order equation $y'' + P(x)y' + Q(x)y = g(x)$, what is the variation of parameters formula for y_p?

a. $y_p = -y_1 \int \frac{y_2 g}{W} dx + y_2 \int \frac{y_1 g}{W} dx$ ✓

b. $y_p = y_1 \int \frac{y_2 g}{W} dx - y_2 \int \frac{y_1 g}{W} dx$

c. $y_p = \int \frac{g(x)}{W} dx$

d. $y_p = -y_2 \int \frac{y_1}{W} dx$

Setting $y_p = u_1 y_1 + u_2 y_2$ yields $u_1' = -\frac{y_2 g}{W}$ and $u_2' = \frac{y_1 g}{W}$, leading directly to $y_p = -y_1 \int \frac{y_2 g}{W} dx + y_2 \int \frac{y_1 g}{W} dx$.

25 [Foundational] • Target: The Variation of Parameters Formula

What must the coefficient of y'' be before applying the variation of parameters formula?

a. **Exactly 1 (must be in standard monic form)** ✓

b. Any non-zero constant

c. Zero

d. Positive

The standard derivation requires the leading coefficient to be normalized to 1; otherwise $g(x)$ must be divided by the coefficient of y''.

26 [Foundational] • Target: The Variation of Parameters Formula

For $y'' + y = \sec x$, where $y_1 = \cos x, y_2 = \sin x$, what is $u_1' = -\frac{y_2 g}{W}$?

a. $-\tan x$ ✓

b. $-\sin x$

c. -1

d. $-\cos x$

With $W = 1$ and $g = \sec x$: $u_1' = -\frac{\sin x \sec x}{1} = -\frac{\sin x}{\cos x} = -\tan x$.

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27 [Foundational] • Target: Solving Advanced Nonhomogeneous Equations

Find $u_1 = \int(-\tan x)dx$ for the problem $y'' + y = \sec x$.

a. $\ln|\cos x|$ ✓

b. $-\ln|\cos x|$

c. $\sec^2 x$

d. $-\cos x$

$u_1 = -\int \tan x dx = -(-\ln|\cos x|) = \ln|\cos x|$.

28 [Foundational] • Target: Solving Advanced Nonhomogeneous Equations

Find $u_2' = \frac{y_1 g}{W}$ for $y'' + y = \sec x$ and evaluate u_2.

a. $u_2 = x$ ✓

b. $u_2 = \sin x$

c. $u_2 = \cos x$

d. $u_2 = \tan x$

$u_2' = \frac{\cos x \sec x}{1} = 1 \Rightarrow u_2 = \int 1 dx = x$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Solving Advanced Nonhomogeneous Equations

Assemble the particular solution $y_p = u_1 y_1 + u_2 y_2$ for $y'' + y = \sec x$.

a. $y_p = \cos x \ln|\cos x| + x \sin x$ ✓

b. $y_p = \cos x + x \sin x$

c. $y_p = \ln|\cos x| + x$

d. $y_p = \sin x \ln|\cos x| + x \cos x$

Combining $u_1 = \ln|\cos x|$ and $u_2 = x$: $y_p = u_1 y_1 + u_2 y_2 = \cos x \ln|\cos x| + x \sin x$.

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30 [Intermediate] • Target: Comparison of Nonhomogeneous Methods

What is the major advantage of Variation of Parameters over Undetermined Coefficients?

a. It works for any continuous forcing function $g(x)$, not just polynomials, exponentials, or sines/cosines ✓

b. It never requires any integration

c. It works without knowing the homogeneous solution y_c

d. It always produces algebraic rather than transcendental answers

Variation of Parameters is completely general and applies to any continuous $g(x)$ (such as $\sec x, \tan x, 1/x$), provided y_c is known.

31 [Intermediate] • Target: Comparison of Nonhomogeneous Methods

What is the main drawback of Variation of Parameters compared to Undetermined Coefficients?

a. The integrals $\int \frac{y_i g}{W} dx$ may be difficult or impossible to evaluate in closed elementary form ✓

b. It only works on first-order equations

c. It gives only approximate numerical solutions

d. It cannot handle initial value problems

Variation of Parameters often produces non-elementary integrals that cannot be expressed using standard functions.

32 [Intermediate] • Target: Comparison of Nonhomogeneous Methods

What is Abel's formula for the Wronskian of solutions to $y'' + P(x)y' + Q(x)y = 0$?

a. $W(x) = Ce^{-\int P(x)dx}$ ✓

b. $W(x) = Ce^{\int Q(x)dx}$

c. $W(x) = C(P(x) + Q(x))$

d. $W(x) = Cx$

Abel's Identity states that the Wronskian satisfies the first-order differential equation $W' + P(x)W = 0$, giving $W(x) = Ce^{-\int P(x)dx}$.

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33 [Intermediate] • Target: Comparison of Nonhomogeneous Methods

Consider the second-order linear nonhomogeneous differential equation $y'' - y = \sin x$. If $y_p(x) = A\cos x + B\sin x$ is a particular solution, find the values of the constants A and B.

a. $A = 0, B = \frac{1}{2}$

b. $A = -\frac{1}{2}, B = 0$

c. $A = 0, B = -\frac{1}{2}$ ✓

d. $A = \frac{1}{2}, B = 0$

We are given the differential equation $y'' - y = \sin x$ and a proposed particular solution $y_p(x) = A\cos x + B\sin x$. We need to find the derivatives of $y_p(x)$ and substitute them into the differential equation. First derivative: $y_p'(x) = -A\sin x + B\cos x$. Second derivative: $y_p''(x) = -A\cos x - B\sin x$. Substitute these into the differential equation: $(-A\cos x - B\sin x) - (A\cos x + B\sin x) = \sin x$ Combine like terms: $(-A - A)\cos x + (-B - B)\sin x = \sin x$.

34 [Intermediate] • Target: The Wronskian

Solve $y'' + 3y' + 2y = -6e^{-4x} - 2$, $y(0) = -1$, $y'(0) = 1$. Find $y(1)$.

a. $-1 - e^{-1} - e^{-4} + \frac{2}{e^2}$ ✓

b. $-\frac{2}{e^2} + e^{-4} + e^{-1} + 1$

c. $-2 - \frac{2}{e} - \frac{2}{e^4} + \frac{4}{e^2}$

d. $-\frac{1}{2} - \frac{1}{2e} - \frac{1}{2e^4} + e^{-2}$

Homogeneous: $r^2 + 3r + 2 = 0$ has roots $r = -2,\ -1$, so $y_c = C_1e^{-2x} + C_2e^{-x}$. Split the forcing (the superposition note): for $-6e^{-4x}$, -4 isn't a root, trial Ae^{-4x}; for -2, trial $Bx + C$ (or just C). Adding the two pieces: $y_p = -1 - e^{-4x}$. With $C_1 = 2, C_2 = -1$ (matching the stated initial conditions), $y = y_c + y_p$, and $y(1)$ follows by substitution.

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35 [Intermediate] • Target: The variation of parameters formula

Solve $y'' - 2y' - 3y = -21e^{-4x} - 3$, $y(0) = -4$, $y'(0) = 0$. Find $y(1)$.

a. $-1 + e^{-4} + \frac{2}{e} + 2e^3$

b. $-4e^3 - \frac{4}{e} - \frac{2}{e^4} + 2$

c. $-e^3 - e^{-1} - \frac{1}{2e^4} + \frac{1}{2}$

d. $-2e^3 - \frac{2}{e} - e^{-4} + 1$ ✓

Homogeneous: $r^2 - 2r - 3 = 0$ has roots $r = 3, -1$, so $y_c = C_1e^{3x} + C_2e^{-x}$. Split the forcing (the superposition note): for $-21e^{-4x}$, -4 isn't a root, trial Ae^{-4x}; for -3, trial $Bx + C$ (or just C). Adding the two pieces: $y_p = 1 - e^{-4x}$. With $C_1 = -2$, $C_2 = -2$ (matching the stated initial conditions), $y = y_c + y_p$, and $y(1)$ follows by substitution.

36 [Intermediate] • Target: A forcing term with no undetermined-coefficients trial form

Solve $y'' + 4y' + 4y = e^{-2x}$, $y(0) = 0$, $y'(0) = 0$. Find $y(2)$.

a. $2e^4$

b. e^{-4}

c. $4e^{-4}$

d. $2e^{-4}$ ✓

Homogeneous: $r^2 + 4r + 4 = (r + 2)^2 = 0 \Rightarrow r = -2$ (repeated), $y_c = (C_1 + C_2x)e^{-2x}$. The forcing e^{-2x} matches the repeated root, so the trial needs x^2: $y_p = Ax^2e^{-2x}$. Substituting gives $2Ae^{-2x} = e^{-2x} \Rightarrow A = \frac{1}{2}$, so $y_p = 12x^2e^{-}2x$. With both conditions zero, $C_1 = C_2 = 0$. So $y = 12x^2e^{-}2x$, and $y(2) = \frac{1}{2}(4)e^{-4} = 2e^{-4}$.

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37 [Intermediate] • Target: A repeated root, with forcing dividing by x

Solve $y'' - y' - 6y = -6e^{-3x} - 6$, $y(0) = -4$, $y'(0) = -4$. Find $y(1)$.

a. $-1 + e^{-3} + e^{-2} + 3e^{3}$

b. $-3e^{3} - e^{-2} - e^{-3} + 1$ ✓

c. $-6e^{3} - \frac{2}{e^{2}} - \frac{2}{e^{3}} + 2$

d. $-\frac{3e^{3}}{2} - \frac{1}{2e^{2}} - \frac{1}{2e^{3}} + \frac{1}{2}$

Homogeneous: $r^2 - r - 6 = 0$ has roots $r = 3, -2$, so $y_c = C_1 e^{3x} + C_2 e^{-2x}$. Split the forcing (the superposition note): for $-6e^{-3x}$, -3 isn't a root, trial Ae^{-3x}; for -6, trial $Bx + C$ (or just C). Adding the two pieces: $y_p = 1 - e^{-3x}$. With $C_1 = -3$, $C_2 = -1$ (matching the stated initial conditions), $y = y_c + y_p$, and $y(1)$ follows by substitution.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Comparing with undetermined coefficients • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Solve $y'' - y' - 2y = -10e^{4x} - 4x$, $y(0) = -7$, $y'(0) = -6$. Find $y(1)$.

a. $-e^{4} - 3e^{2} - \frac{2}{e} + 1$ ✓

b. $-1 + \frac{2}{e} + 3e^{2} + e^{4}$

c. $-2e^{4} - 6e^{2} - \frac{4}{e} + 2$

d. $-\frac{e^{4}}{2} - \frac{3e^{2}}{2} - e^{-1} + \frac{1}{2}$

Homogeneous: $r^2 - r - 2 = 0$ has roots $r = -1, 2$, so $y_c = C_1 e^{-x} + C_2 e^{2x}$. Split the forcing (the superposition note): for $-10e^{4x}$, 4 isn't a root, trial Ae^{4x}; for $-4x$, trial $Bx + C$ (or just C). Adding the two pieces: $y_p = 2x - e^{4x} - 1$. With $C_1 = -2$, $C_2 = -3$ (matching the stated initial conditions), $y = y_c + y_p$, and $y(1)$ follows by substitution.

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39 [Challenging] • Target: When to prefer each method • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

Solve $y'' + 2y' - 3y = -21e^{4x} - 9x$, $y(0) = 2$, $y'(0) = 8$. Find $y(1)$.

a. $-3e - 5 + \frac{2}{e^3} + e^4$

b. $-2e^4 - \frac{4}{e^3} + 10 + 6e$

c. $-\frac{e^4}{2} - e^{-3} + \frac{5}{2} + \frac{3e}{2}$

d. $-e^4 - \frac{2}{e^3} + 5 + 3e$ ✓

Homogeneous: $r^2 + 2r - 3 = 0$ has roots $r = -3, 1$, so $y_c = C_1e^{-3x} + C_2e^{x}$. Split the forcing (the superposition note): for $-21e^{4x}$, 4 isn't a root, trial Ae^{4x}; for $-9x$, trial $Bx + C$ (or just C). Adding the two pieces: $y_p = 3x - e^{4x} + 2$. With $C_1 = -2$, $C_2 = 3$ (matching the stated initial conditions), $y = y_c + y_p$, and $y(1)$ follows by substitution.

40 [Challenging] • Target: Solving $y''+y=\csc x$ in full • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

Solve $y'' - y' - 6y = 6e^{-3x} + 6$, $y(0) = 2$, $y'(0) = -12$. Find $y(1)$.

a. $-e^3 - 1 + e^{-3} + \frac{3}{e^2}$ ✓

b. $-\frac{3}{e^2} - e^{-3} + 1 + e^3$

c. $-2e^3 - 2 + \frac{2}{e^3} + \frac{6}{e^2}$

d. $-\frac{e^3}{2} - \frac{1}{2} + \frac{1}{2e^3} + \frac{3}{2e^2}$

Homogeneous: $r^2 - r - 6 = 0$ has roots $r = -2, 3$, so $y_c = C_1e^{-2x} + C_2e^{3x}$. Split the forcing (the superposition note): for $6e^{-3x}$, -3 isn't a root, trial Ae^{-3x}; for 6, trial $Bx + C$ (or just C). Adding the two pieces: $y_p = -1 + e^{-3x}$. With $C_1 = 3$, $C_2 = -1$ (matching the stated initial conditions), $y = y_c + y_p$, and $y(1)$ follows by substitution.

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41 [Challenging] • Target: Variation of parameters with a shifted independent variable • Cross-Lesson: 1.01 Separable Equations

Solve $y'' - y' - 2y = -10e^{-3x} - 2$, $y(0) = 2$, $y'(0) = 4$. Find $y(1)$.

a. $-e^2 - 1 - e^{-1} + e^{-3}$

b. $-\frac{1}{e^3} + e^{-1} + 1 + e^2$ ✓

c. $-\frac{2}{e^3} + \frac{2}{e} + 2 + 2e^2$

d. $-\frac{1}{2e^3} + \frac{1}{2e} + \frac{1}{2} + \frac{e^2}{2}$

Homogeneous: $r^2 - r - 2 = 0$ has roots $r = -1, 2$, so $y_c = C_1e^{-x} + C_2e^{2x}$. Split the forcing (the superposition note): for $-10e^{-3x}$, -3 isn't a root, trial Ae^{-3x}; for -2, trial $Bx + C$ (or just C). Adding the two pieces: $y_p = 1 - e^{-3x}$. With $C_1 = 1$, $C_2 = 1$ (matching the stated initial conditions), $y = y_c + y_p$, and $y(1)$ follows by substitution.

42 [Challenging] • Target: A summary comparison • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Solve $y'' - y = -15e^{-4x} - 1x$, $y(0) = -2$, $y'(0) = 0$. Find $y(1)$.

a. $-3e - e^{-4} + \frac{2}{e} + 1$ ✓

b. $-1 - \frac{2}{e} + e^{-4} + 3e$

c. $-6e - \frac{2}{e^4} + \frac{4}{e} + 2$

d. $-\frac{3e}{2} - \frac{1}{2e^4} + e^{-1} + \frac{1}{2}$

Homogeneous: $r^2 - 1 = 0$ has roots $r = 1, -1$, so $y_c = C_1e^{x} + C_2e^{-x}$. Split the forcing (the superposition note): for $-15e^{-4x}$, -4 isn't a root, trial Ae^{-4x}; for $-1x$, trial $Bx + C$ (or just C). Adding the two pieces: $y_p = x - e^{-4x}$. With $C_1 = -3$, $C_2 = 2$ (matching the stated initial conditions), $y = y_c + y_p$, and $y(1)$ follows by substitution.

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43 [Challenging] • Target: The Wronskian Determinant • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

Solve $y'' - y = -15e^{-4x} - 1$, $y(0) = 0$, $y'(0) = 2$. Find $y(1)$.

a. $-1 - e^{-1} + e^{-4} + e$

b. $-e - e^{-4} + e^{-1} + 1$ ✓

c. $-2e - \frac{2}{e^4} + \frac{2}{e} + 2$

d. $-\frac{e}{2} - \frac{1}{2e^4} + \frac{1}{2e} + \frac{1}{2}$

Homogeneous: $r^2 - 1 = 0$ has roots $r = -1, 1$, so $y_c = C_1e^{-x} + C_2e^{x}$. Split the forcing (the superposition note): for $-15e^{-4x}$, -4 isn't a root, trial Ae^{-4x}; for -1, trial $Bx + C$ (or just C). Adding the two pieces: $y_p = 1 - e^{-4x}$. With $C_1 = 1$, $C_2 = -1$ (matching the stated initial conditions), $y = y_c + y_p$, and $y(1)$ follows by substitution.

44 [Challenging] • Target: The Variation of Parameters Formula • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

Solve $y'' + 10y' + 25y = 3e^{-5x}$, $y(0) = 3$, $y'(0) = -13$. Find $y(1)$.

a. $\frac{13}{2e^5}$ ✓

b. $-\frac{13}{2e^5}$

c. $\frac{13}{e^5}$

d. $\frac{13}{4e^5}$

Homogeneous: $r^2 + 10r + 25 = 0 \Rightarrow (r + 5)^2 = 0$ has roots $r = -5$ (repeated), so $y_c = (C_1 + C_2x)e^{-5x}$. Forcing $3e^{-5x}$ matches the repeated root $r = -5$ (double resonance): trial $y_p = Ax^2e^{-5x}$ gives $y_p = \frac{3x^2e^{-5x}}{2}$. With $C_1 = 3$, $C_2 = 2$ (matching the stated initial conditions), $y = y_c + y_p$, and $y(1)$ follows by substitution.

45 [Challenging] • Target: Solving Advanced Nonhomogeneous Equations • Cross-Lesson: 1.01 Separable Equations

Solve $y'' + 8y' + 16y = 3e^{-4x}$, $y(0) = -2$, $y'(0) = 11$. Find $y(2)$.

a. $\frac{10}{e^8}$ ✓

b. $-\frac{10}{e^8}$

c. $\frac{20}{e^8}$

d. $\frac{5}{e^8}$

Homogeneous: $r^2 + 8r + 16 = 0 \Rightarrow (r + 4)^2 = 0$ has roots $r = -4$ (repeated), so $y_c = (C_1 + C_2x)e^{-4x}$. Forcing $3e^{-4x}$ matches the repeated root $r = -4$ (double resonance): trial $y_p = Ax^2e^{-4x}$ gives $y_p = \frac{3x^2e^{-4x}}{2}$. With $C_1 = -2$, $C_2 = 3$ (matching the stated initial conditions), $y = y_c + y_p$, and $y(2)$ follows by substitution.

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